



Module 2: Propagation & Practices

Advanced Propagation Modes · Test Equipment · Receiver Performance
2024-2028 Amateur Extra Question Pool



What We're Covering Today

- 8 exam questions — E3 (3 Qs) + E4 (5 Qs) = 16% of your Extra exam
- Hour 1: Advanced propagation — modes you won't see on General
 - → EME, meteor scatter, auroral, tropospheric ducting
 - → TEP, long-path, sporadic E, chordal hop
 - → Space weather: solar flux, A/K index, Bz, VOACAP
- Hour 2: Amateur practices
 - → Test equipment: oscilloscope, spectrum analyzer, antenna analyzer, VNA
 - → Measurement: S-parameters, directional wattmeter, Q measurement
 - → Receiver performance: noise, dynamic range, blocking, SDR overload
 - → Noise and interference sources



Part 1: EME, Meteor Scatter & Auroral

Subelement E3A — 1 exam question



Earth-Moon-Earth (EME / Moon Bounce)

- The Moon reflects VHF/UHF signals — you can work stations on the other side of the world
- Maximum one-way path: ~12,000 miles (Earth to Moon and back = ~480,000 miles round trip)
- Perigee (Moon closest to Earth): minimum path loss, strongest signals
- Libration fading: Moon wobbles slightly (libration) — antenna orientation constantly shifts
- → Causes rapid, irregular signal fluctuations — 'fluttery' audio
- Mode: Q65 — designed for severe Doppler shifting and libration fading
- Requires very high-gain antennas (big Yagi arrays) and usually maximum legal power



Meteor Scatter

- Meteors burn up in the E region of the ionosphere (~80-120 km altitude)
- The ionized trail reflects VHF signals — for milliseconds to a few seconds
- Frequency range: 28-148 MHz (10m through 2m)
- Mode: MSK144 — designed for very short burst duration
- MSK144 encodes data quickly enough to fit a full exchange in a single burst
- Peak meteor scatter activity: major showers (Perseids in August, Leonids in November)
- Random (sporadic) meteors: year-round low-level activity



Auroral and Tropospheric Ducting

- Auroral propagation: severe geomagnetic storms excite the auroral oval
- → The aurora acts as a reflector for VHF signals
- → CW is the only practical mode — auroral reflection distorts audio badly
- → Signals have a characteristic 'buzz' or 'hiss' quality
- → Works best when you point your antenna toward the aurora (northward in N. hemisphere)

- Microwave tropospheric ducting:
- → Temperature inversions over large bodies of water trap signals
- → Typical range: 100-300 miles; sometimes much farther
- → Most common on microwave bands (2.4 GHz and above)
- → Florida to Cuba; California to Hawaii (rare but happens)



EM Wave Fundamentals

- An EM wave has two fields at right angles to each other: E (electric) and H (magnetic)
- The wave travels at right angles to BOTH E and H fields
- Think of it as three perpendicular axes: travel direction, E field, H field
- Circular polarization: both E and H fields rotate as the wave propagates
- → Reduces spin modulation on satellites (covered in Module 1)
- Speed through a medium: determined by the index of refraction
- → Slower in denser media (glass, water, ionosphere) than in vacuum
- MUF at night: as ionosphere thins at darkness, MUF drops — switch to a lower HF band



Part 2: TEP, Long-Path, Sporadic E

Subelement E3B — 1 exam question



Transequatorial Propagation (TEP)

- TEP enables 2,000-3,000 mile paths perpendicular to the geomagnetic equator
- Maximum range: approximately 5,000 miles
- Both stations must be roughly equidistant north and south of the geomagnetic equator
- Timing: typically afternoon to early evening
- This is a 2m DX mode — it enables contacts that are otherwise impossible
- Not the same as standard F-layer propagation — TEP uses field-aligned irregularities
- Example: Florida to southern South America on 144 MHz



Long-Path, 160m, and Chordal Hop

- Long-path propagation: signals travel the 'long way' around Earth (>180 degrees of arc)
- → Most frequent on 40m and 20m
- → Point antenna in the OPPOSITE direction from normal short-path
- → When short-path is blocked or noisy, long-path sometimes gives a better signal
- 160m (1.8 MHz) propagation:
 - → Path must be ENTIRELY in darkness — even a short illuminated segment causes absorption
 - → Best windows: around your local sunrise/sunset
- Chordal hop: successive ionospheric refractions without ground touching between hops
 - → Less loss than standard hops (no ground absorption)
 - → Lower elevation angle = longer individual hop distance



Sporadic E and Ground Wave

- Sporadic E (Es): random patches of intense ionization in the E layer
- → Peak occurrence: around solstices, especially summer (June-July N. hemisphere)
- → Occurs most often between sunrise and sunset
- → Can enable DX on 6m, 2m, 10m without F-layer support
- → Unpredictable — check DX clusters and spotting networks

- Ordinary and extraordinary waves:
 - → Two waves that propagate independently through the ionosphere
 - → Both are elliptically polarized

- Ground wave: travels along Earth's surface
 - → Uses vertical polarization
 - → Range decreases as frequency increases



Part 3: Space Weather

Subelement E3C — 1 exam question



Space Weather Indices

Solar Flux Index (SFI)	UV/EUV from Sun; higher = better HF propagation
A-index	Daily geomagnetic average (0-400); rising = worse HF
K-index	3-hour geomagnetic reading (0-9); K>4 = disrupted HF
Bz (north-south IMF)	Southward Bz = more geomagnetic disturbance
G-scale storms	G1 (minor) through G5 (extreme)
Solar flare class X	Most intense class; causes HF radio blackouts
304A parameter	UV emissions; correlated with solar flux index
VOACAP	HF propagation prediction software
Auroral oval when A/K high	High absorption on polar paths; auroral scatter on VHF
Sudden HF noise increase	CME impact or solar flare in progress
VHF radio horizon	~15% farther than geographic horizon



Part 4: Test Equipment

Subelement E4A — 1 exam question



Oscilloscope and Spectrum Analyzer

- Digital oscilloscope:
 - → Displays voltage vs. TIME
 - → Sampling rate limits the highest frequency that can be accurately displayed
 - → Aliasing: if signal frequency > half the sampling rate, it appears at a false lower frequency
 - → Probe compensation: feed a square wave, adjust trimmer for flat top (no overshoot)
 - → Line trigger (AC power line sync): best mode for measuring power supply ripple
- Spectrum analyzer:
 - → Displays AMPLITUDE vs. FREQUENCY
 - → Vertical axis = amplitude (usually in dBm)
 - → Horizontal axis = frequency
 - → Best tool for viewing spurious emissions, harmonics, and IMD products



Antenna Analyzer and Frequency Counter

- Antenna analyzer:
 - → Computes SWR and feed point impedance
 - → Can measure velocity factor of coax (from electrical length vs. physical length)
 - → Can find cable length by measuring resonant frequency of a shorted stub
 - → Can find resonant frequency of an antenna
- Frequency counter:
 - → Measures frequency by counting cycles over a precise time interval
 - → Accuracy is determined by the time base (reference oscillator) accuracy
 - → Better oscillator = more accurate counter
- Prescaler:
 - → Reduces a high-frequency signal to within the counter's operating range



VNA and Measurement Techniques (E4B)

- VNA (Vector Network Analyzer): measures S-parameters of RF circuits
 - → Calibration: short circuit + open circuit + 50-ohm load — establishes reference planes
 - → Two-port VNA: measures filter frequency response
 - S_{11} = input return loss / input VSWR (reflection at input port)
 - S_{21} = forward gain (signal from port 1 to port 2)
- Directional wattmeter: measures forward and reflected power
 - → Power absorbed = forward power - reflected power
 - → Example: 100W forward, 25W reflected: 75W absorbed
- Q measurement: $Q = \text{resonant frequency} / 3\text{-dB bandwidth}$
- IMD measurement: inject two non-harmonically related tones; view IMD products on spectrum



Part 5: Receiver Performance

Subelements E4C-D — 1 exam question



Noise and Sensitivity

- Thermal noise floor: -174 dBm in 1 Hz bandwidth at room temperature
- Noise figure (NF): how much the receiver adds above the thermal minimum
- MDS (Minimum Discernible Signal): weakest detectable signal in the receiver
- Bandwidth increases noise floor:
 - → 50 Hz to 1000 Hz: $10 \times \log_{10}(1000/50) = +13$ dB noise floor increase
 - → Narrower filter = lower noise floor = weaker signals detectable
- Phase noise in SDR master clock: LO phase noise mixes with nearby strong signals
 - → Creates in-band interference that looks like receiver noise
- Preselector/front-end filter: bandpass filter eliminating strong out-of-band signals before the LNA



Dynamic Range and Blocking

- Blocking dynamic range: range from noise floor to signal level causing 1 dB gain compression
- → A strong nearby signal compresses the front end — nearby signals get weaker
- Third-order intercept point (IP3):
- → Theoretical signal level where IMD products equal the input signal
- → Higher IP3 = better receiver in crowded band conditions
- Reciprocal mixing: LO phase noise mixes with adjacent strong signal → creates interference
- Roofing filter: wide crystal filter at first IF — reduces reciprocal mixing effects
- High IF frequency: image is farther away — easier to reject
- SDR overload: input signal exceeds ADC reference voltage → add attenuation
- Desensitization: strong nearby signal compresses front-end gain
- → Fix: insert attenuation before the first RF stage



Noise and Interference Sources (E4E)

Noise Sources

- Switch-mode power supplies: broadband noise at regular frequency intervals across HF
- Power line noise: buzzing, worst on low HF
- Corroded metal: rusty bolt effect — mixes and reradiates broadcast signals
- Ignition noise: impulse noise from vehicles
- Common-mode current: flows equally on all conductors

Mitigations

- ANF (Auto Notch Filter): removes carriers and CW — but also removes a desired CW signal
- Noise Blanker: impulse noise; can distort strong signals
- Digital NR: broadband and power line noise
- Ferrite choke on power cables: blocks common-mode
- Single point ground panel: all protectors activate simultaneously



Checkpoint Quiz 1 / 10

What type of fading is associated with EME propagation?

- A. Multipath fading
- B. Libration fading
- C. Rayleigh fading
- D. Selective fading

Answer: B — Libration fading: Moon's wobble causes rapid irregular signal variations



Checkpoint Quiz 2 / 10

Which mode is best suited for meteor scatter propagation?

- A. FT8
- B. Q65
- C. MSK144
- D. WSPR

Answer: C — MSK144 is designed for the very short burst duration of meteor scatter



Checkpoint Quiz 3 / 10

Why is CW preferred for auroral propagation?

- A. CW requires less power
- B. Auroral reflection distorts audio — phone signals are unintelligible
- C. The aurora only reflects CW frequencies
- D. CW has a narrower beamwidth

Answer: B — Auroral reflection distorts audio; CW tone is still copyable through the distortion



Checkpoint Quiz 4 / 10

What does a southward Bz indicate?

- A. Improved HF propagation
- B. Enhanced sporadic E
- C. Increasing geomagnetic disturbance
- D. Higher solar flux

Answer: C — Southward Bz couples energy into the magnetosphere, increasing disturbance



Checkpoint Quiz 5 / 10

What does aliasing on a digital oscilloscope look like?

- A. Overshoot on square wave edges
- B. A signal appearing at a false lower frequency
- C. Loss of vertical resolution
- D. Baseline drift

Answer: B — Aliasing: undersampled signal appears at a false (lower) frequency



Checkpoint Quiz 6 / 10

What is the primary factor determining frequency counter accuracy?

- A. Input impedance
- B. Display resolution
- C. Time base accuracy
- D. Prescaler ratio

Answer: C — Counter accuracy is determined by the accuracy of its time base (reference oscillator)



Checkpoint Quiz 7 / 10

In a VNA two-port measurement, what does S21 represent?

- A. Input return loss
- B. Output return loss
- C. Forward gain from port 1 to port 2
- D. Reverse isolation

Answer: C — S21: signal into port 1, measured at port 2 — forward gain



Checkpoint Quiz 8 / 10

A receiver's noise floor increases by how many dB when bandwidth goes from 50 Hz to 1000 Hz?

- A. 10 dB
- B. 3 dB
- C. 20 dB
- D. 13 dB

Answer: D — $10 \times \log_{10}(1000/50) = 10 \times 1.301 = 13 \text{ dB}$



Checkpoint Quiz 9 / 10

What is the rusty bolt effect?

- A. Coax connector corrosion increasing SWR
- B. Corroded metal connections mixing and reradiating broadcast signals
- C. Galvanic corrosion in antenna elements
- D. Moisture in antenna insulators

Answer: B — Corroded metal junctions act as passive mixers, generating spurious signals from nearby broadcast stations



Checkpoint Quiz 10 / 10

Which receiver filter improves blocking dynamic range?

- A. Audio low-pass filter
- B. ANF
- C. Roofing filter
- D. Noise blanker

Answer: C — Roofing filter (crystal filter at first IF) reduces reciprocal mixing and improves blocking dynamic range



Before Module 3

Your assignment:

- Drill HamStudy E3A-E3C and E4A-E4E to 80%+ confidence
- Focus extra time on E4C-D (receiver performance) — most students find it the hardest
- E4B (S-parameters, directional wattmeter): get those formulas locked in

Coming up:

Module 3: Electrical Principles — resonance, Q, time constants, phasors. Only 4 exam questions but critical foundation for Modules 5 and 7. Come with a calculator.